

Beyond this meeting

A key aspect of CoFI is the flexibility provided for exploration and experimentation in inversion

Created examples and brought existing capabilities together

Making unique approaches accessible through the common interface and grow the set of examples

A FAIR example driven framework...

A conversation about CoFI over coffee is just the beginning...



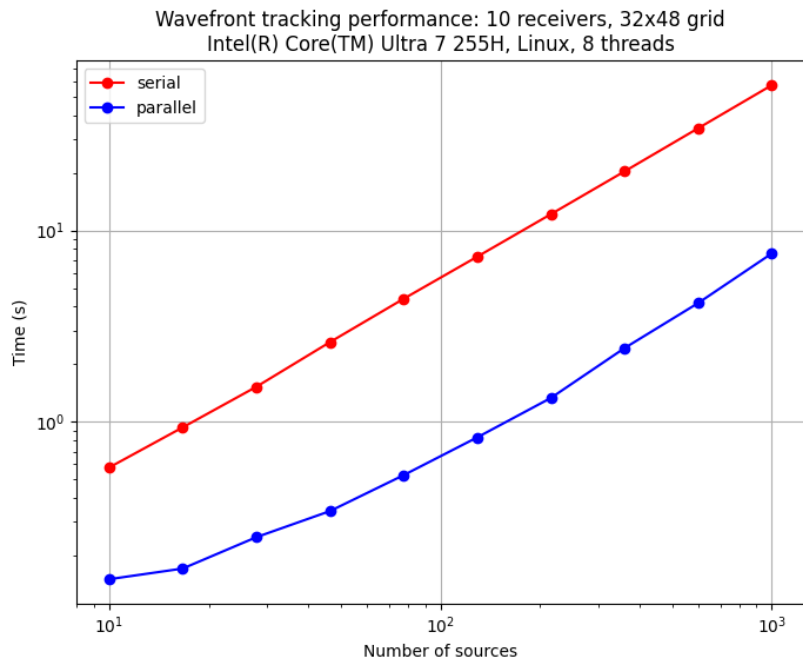
Explore our examples

Setup your own inverse problem

Let us know what you think



Scalable CoFI...

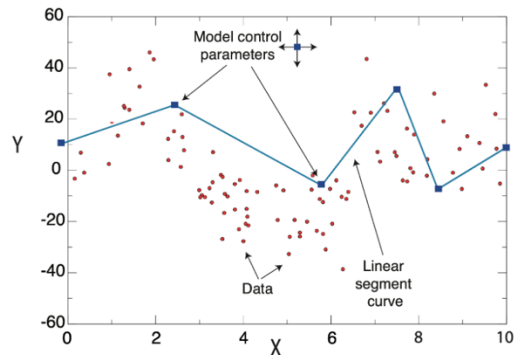


Forward problems are frequently the bottleneck in inverse problem workflows.

In many parameter estimation methods, communication typically occurs via linear algebra steps.

In contrast, ensemble methods often exhibit less communication between walkers, chains, or states

Beyond trans-dimensional sampling



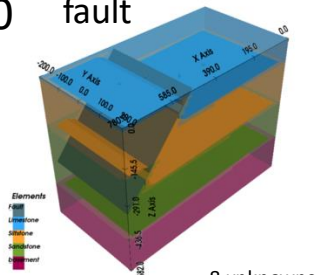
github.com/fmagrini/bayes-bay github.com/inlab-geo/pyrjcmc

Unified Interface for ensemble methods

```
def log_probability(theta, x, y, yerr):
    lp = log_prior(theta)
    if not np.isfinite(lp):
        return -np.inf
    return lp + log_likelihood(theta, x, y, yerr)
```

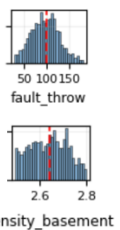
emcee.readthedocs.io/en/stable/

0 fault



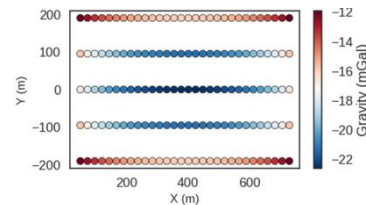
8 unknowns

Trans-conceptual sampling

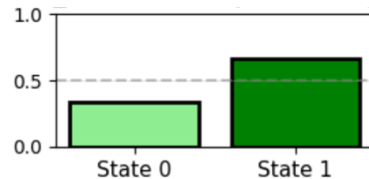


...

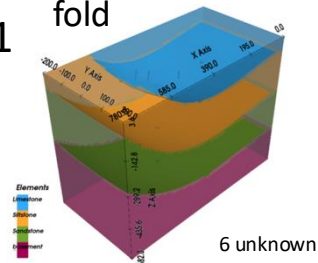
Both states can fit the gravity response of state/model 0



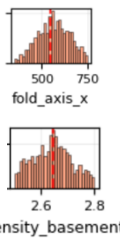
Fits are equal – thus the simpler model is preferred...



1 fold



6 unknowns

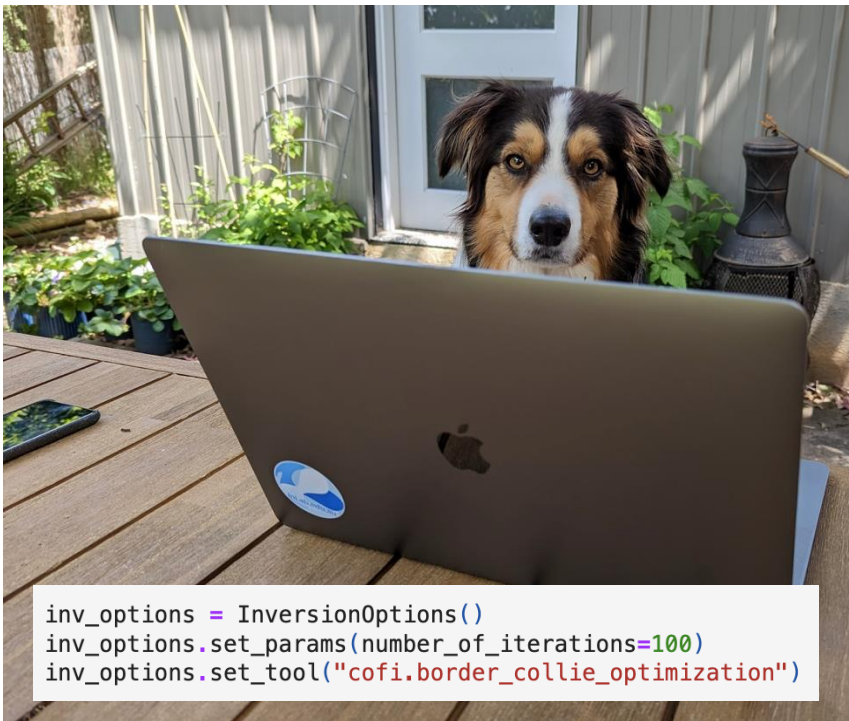


...

github.com/gempy-project/gempy

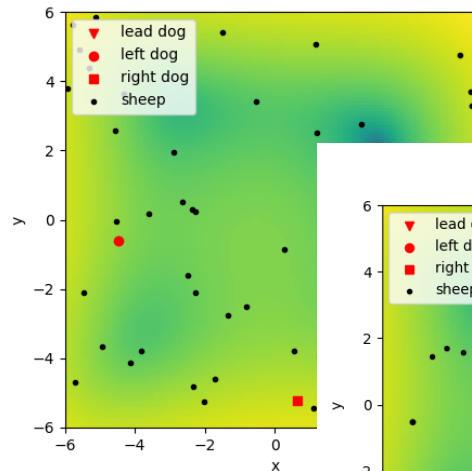
github.com/inlab-geo/pyTransC

But what about my favourite metaheuristic optimiser?

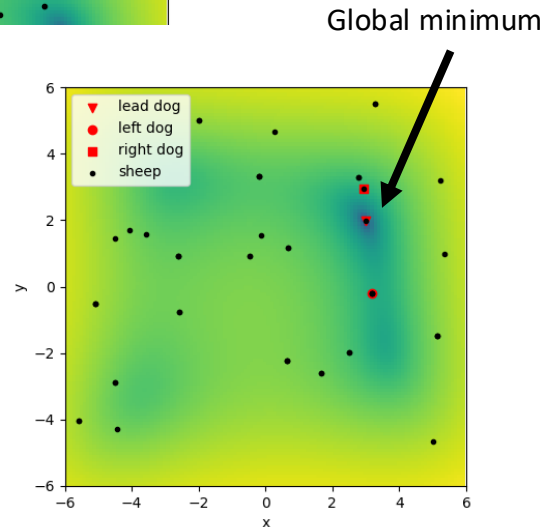


```
inv_options = InversionOptions()  
inv_options.set_params(number_of_iterations=100)  
inv_options.set_tool("cofi.border_collie_optimization")
```

Border Collie Optimization



www.inlab.au



Wrap up

This is an **open source** project

Feedback:

Choosing priorities: Expanding the methods; the examples, building a community;

Wishlist includes: scalability (HPC); new examples; automatic differentiation, Surrogate modelling. Link to new optimization and sampling packages...

Work with us & get involved:

creating an application example using CoFI

join our slack community:

<http://inlab.au/community>

raise an issue on github:

<https://github.com/inlab-geo/cofi/issues>